for various input/output functions. In some meters, most of these modules RTC, LCD controller, Temperature sensor, memory, ADC (Analog Digital Converter) may be implemented inside the processing and communication circuit.



Fig 5.17 *Digital Energy Meter*

Advantages of Digital Energy Meters

i) High accuracy

ii) Robustness

iii) No moving parts

iv) Easy to gauge readings through digital display

v) Over current protection

vi) Readings can be stored and print out may be taken

vii) Smaller in size

viii) Consumes less power

ix) Long life

x) Easy to carry anywhere

xi) Remote control is possible

Points to Remember

- Induced E.M.F $e = Blv \sin\theta$
- Frequency $f = \frac{PN}{120}$ Hz
- Average value $= \frac{2I_m}{\pi}$ or $\frac{2V_m}{\pi}$
- R.M.S. value $= \frac{I_m}{\sqrt{2}}$ or $\frac{V_m}{\sqrt{2}}$
- Form factor $= \frac{\text{RMS Value}}{\text{Average Value}} = 1.11$
- Peak factor $= \frac{\text{Max. Value}}{\text{RMS Value}} = 1.414$
- Power factor $= \frac{V_I \cos\theta}{V_I}$
- In pure resistance circuit, power factor is one (unity)
- In pure inductive or capacitive circuit, power factor is zero
- Inductive reactance $X_L = 2\pi fl$
- Capacitive reactance $X_C = 1/2\pi f_C$
- Impedance $Z = \frac{V}{I}$

In RLC series circuit powerfactor $\cos\theta = \frac{R}{Z}$

Both in RLC series and RLC parallel circuit

- Power $P = VI \cos\theta$ watts

In Star Connection

Phase current = Line current

$$I_{ph} = I_L$$

Phase voltage $= \frac{\text{(Line Voltage)}}{\sqrt{3}}$

$$V_{ph} = \frac{V_L}{\sqrt{3}}$$

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